Derivation of tower scaling

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- Tower base moment: $M \sim TH$
- Rotor thrust scaling: $T \sim P^{\frac{2}{3}}$

Where P is the rated power, H is the hub height. This leads to

$$M \sim P^{2/3} H$$

For geometrically similar tubular tower,

- Tower mass: $m \sim D^3$
- Moment of inertia: $I \sim D^4$

Where D is the tower diameter.

From the bending stress relation $\sigma = \frac{MD}{2I}$, rearrange M with the similarities above:

$$M = 2\sigma \frac{I}{D} \sim D^3 \sim m$$

Finally,

$$m \sim P^{2/3} H$$